Can You Predict the Next California Wildfire?

A Data Science Case Study in Time Series & Machine Learning

It’s January 2025. Fires are tearing through Los Angeles and San Diego.
What if a model had flagged days wildfires would occur weeks in advance?

Your Mission

You are a data scientist at a California emergency preparedness agency. Leadership has handed you decades of daily weather records (temperature, humidity, wind speed, precipitation) alongside a log of wildfire ignition events going back to 1984. They want a model that can flag dangerous days before they happen.

Your job: explore the data, identify the seasonal patterns that precede wildfires, build and compare predictive models, and deliver a recommendation on which approach the agency should deploy.

The Challenge

Wildfires are rare events, which makes prediction hard. Missing a fire has devastating consequences, but too many false alarms and decision-makers stop listening. You’ll need to think carefully about not just accuracy, but which errors matter most.

What You’ll Build

- An exploratory analysis revealing seasonal weather patterns tied to wildfire risk
- A logistic regression baseline and a random forest model trained on 1985–2022 data and tested on 2023
- A performance evaluation focused on recall because missing fires is not an option
- A written recommendation: which model would you deploy, and why?

Skills You’ll Practice

Time series analysis · Logistic regression · Random forests · Class imbalance handling · Model evaluation (precision, recall, F1) · Python (scikit-learn, pandas)

Full specifications, data, starter code, and references are in the GitHub repository. See the rubric for exact deliverable requirements.